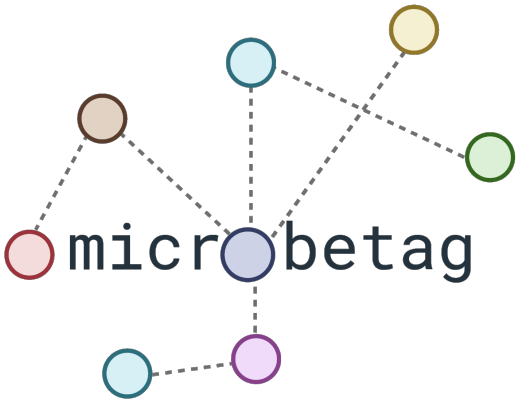
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# Gossip in the Microbiome: Metabolic Secrets Behind Microbial Co-Occurrence

## HSCBB26

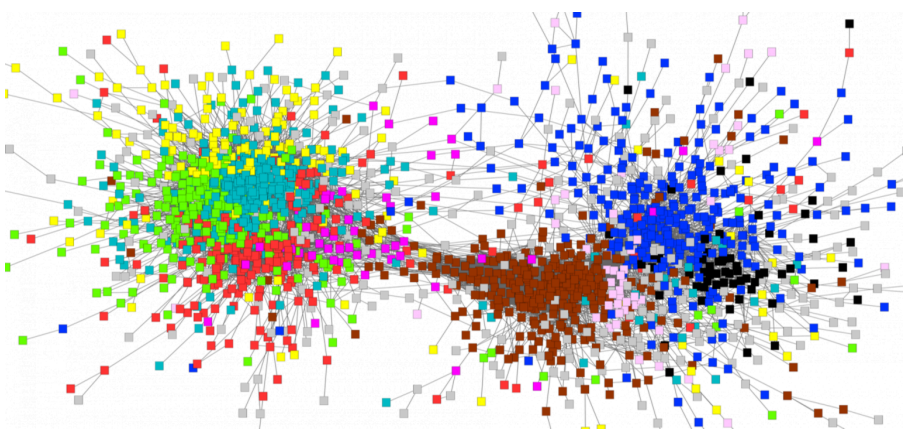
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Mechanisms of Microbial Metabolic Interactions (m3i)
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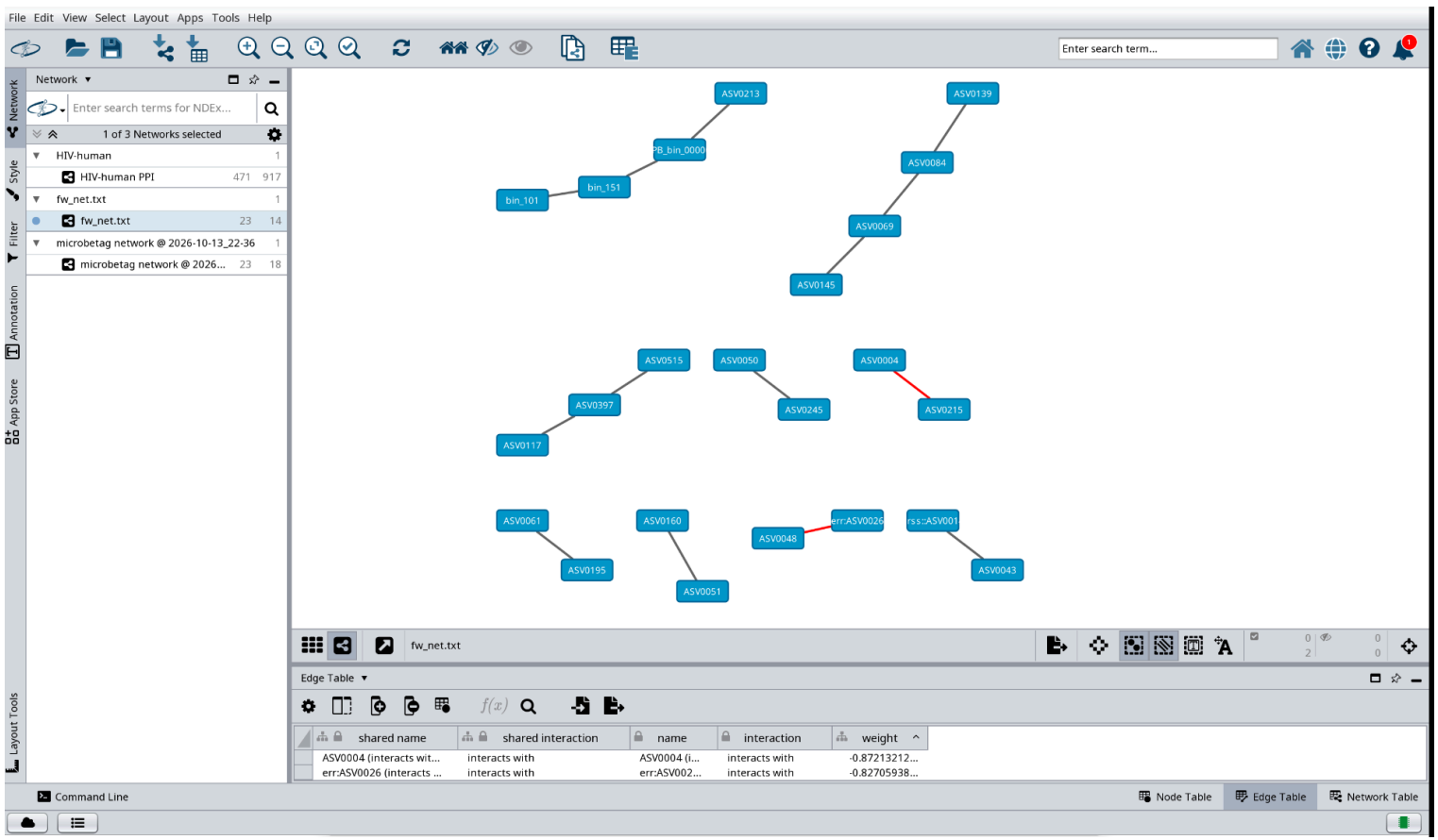
# Making sense of a hairball is hard

Amplicon / metagenomic surveys → **co-occurrence networks**: thousands of **undirected** edges between taxa that appear together (or avoid each other).



*Who* is working with *who* and *how*?

# A co-occurrence network in Cytoscape



# microbetag: a hypothesis generator

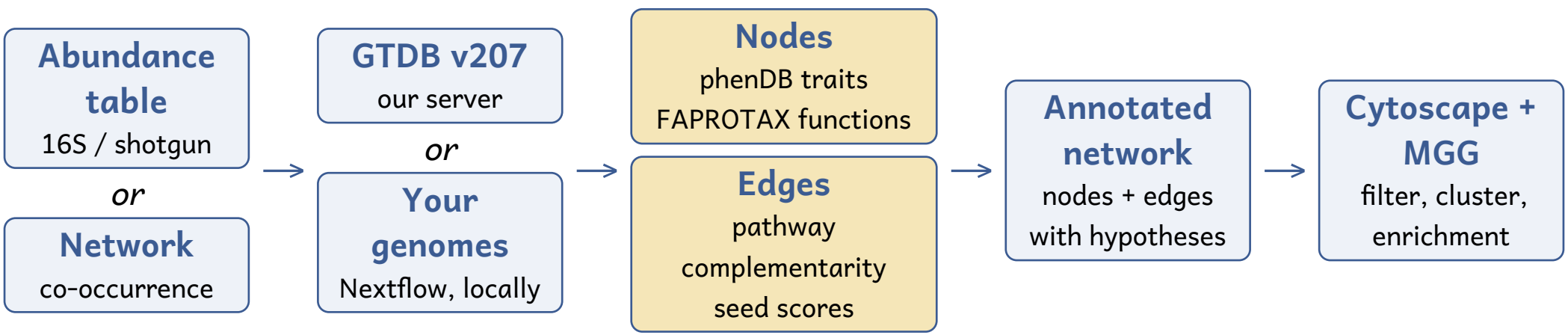
We cannot prove interactions from co-occurrence / mutual exclusion alone, but genomes can suggest **why** two taxa show such a pattern:

- *complementary* pathways → **cross-feedings**?
- overlapping needs → **competition**?
- shared traits → same **niche**?

▹ microbetag

Turns the hairball into **testable hypotheses**

# How microbetag works



# Node level: what are a species' phenotypic traits?

Each taxon gets **functional traits**: genome-based predictions (phenotrex) and literature-based **FAPROTAX** functions (105 in total), e.g.:

▹ Carbon

- methanogenesis
- fermentation
- cellulolysis

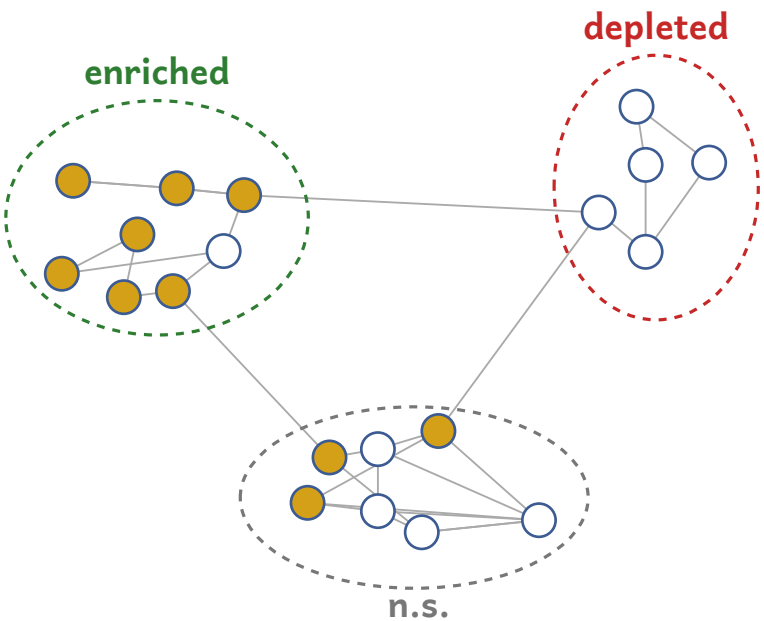
▹ Nitrogen & sulfur

- nitrogen fixation
- denitrification
- sulfate respiration

▹ Lifestyle

- obligate anaerobes
- human gut
- plant pathogen

# Clusters & enrichment *analysis*




- taxon **with** the trait
- taxon **without**

1. **Cluster** the network (e.g. manta)
2. Per cluster & trait: **hypergeometric test** (wiki)
3. FDR correction → **enriched** / **depleted**

Each cluster reads as a **functional guild**

# Edge level: what could they exchange?

▹ Pathway complementarity

KEGG modules one taxon misses and its partner can complete

▹ Seed scores

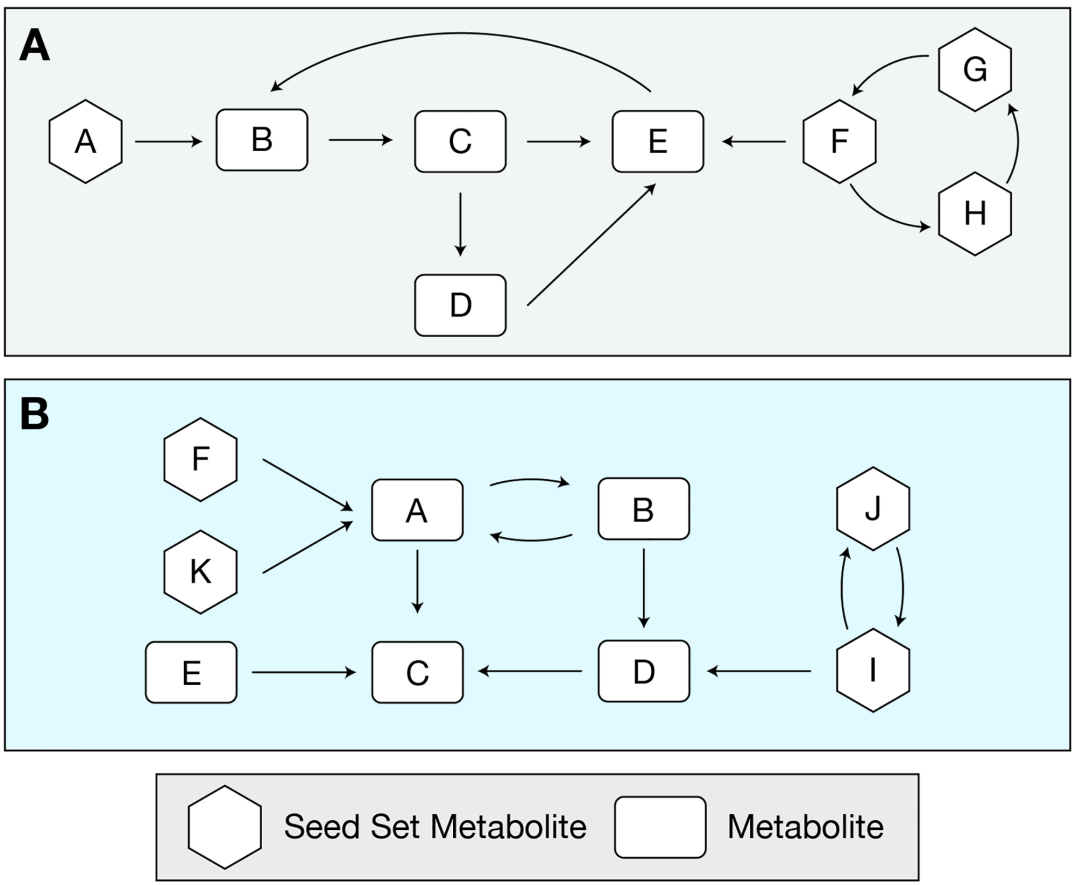
Competition and cooperation potential from metabolic network seed sets

▹ Seed complementarity

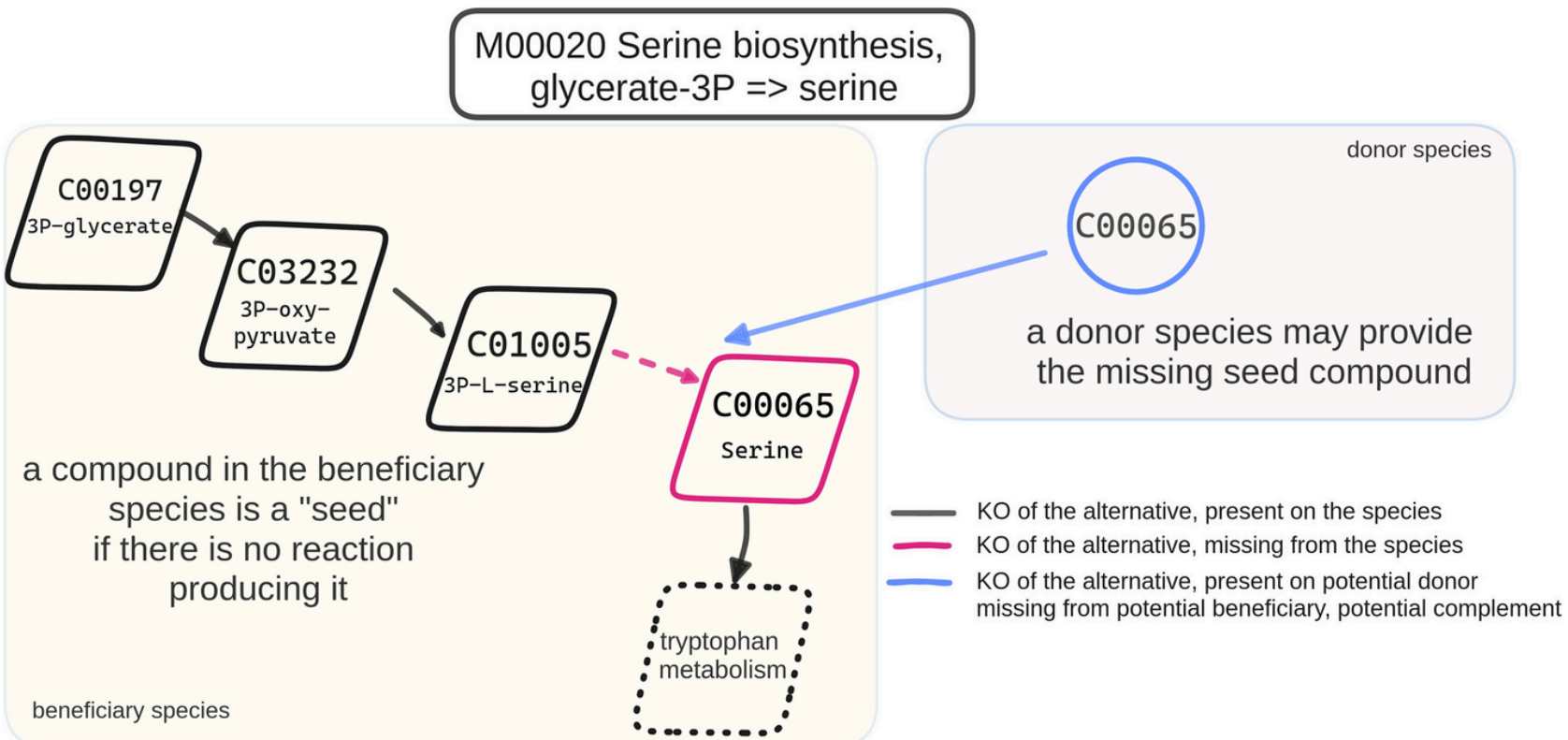
Candidate exchanged compounds from genome-scale models

# The seed concept

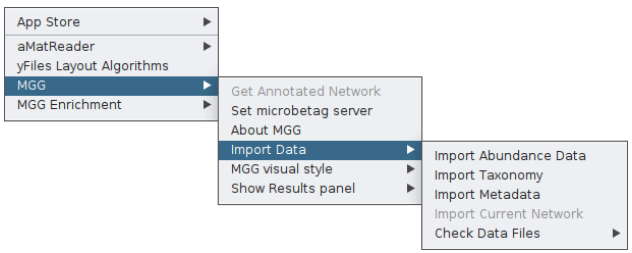
Pseudo metabolitc networks of species A and B



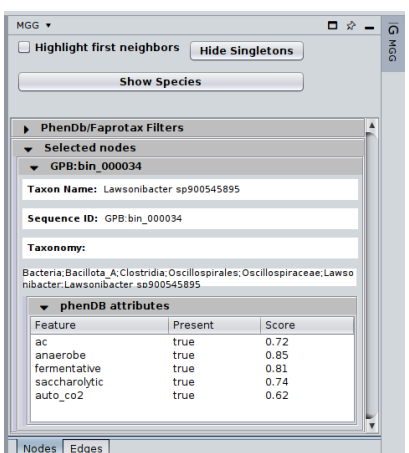
# Seed complementarity



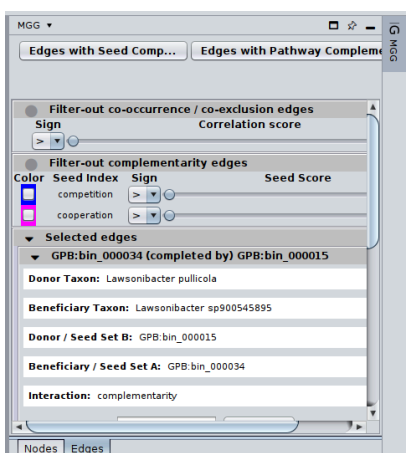
# Parsing the network with MGG



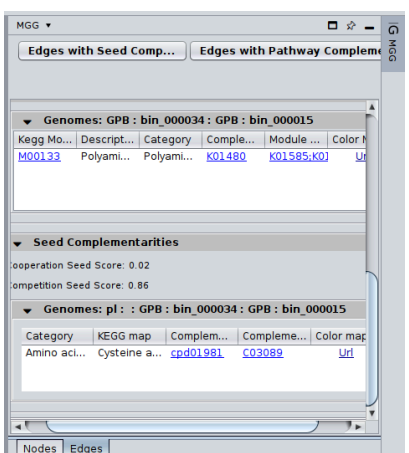

1. **Import** abundance table, taxonomy, metadata
2. **Get annotated network** from our server, or import your local one
3. Click *a* **node** or **edge** to read its annotations, **filter** edges by correlation or seed scores
4. Run **MGG Enrichment** on clusters




**Nodes**: taxonomy, phenDB / FAPROTAX

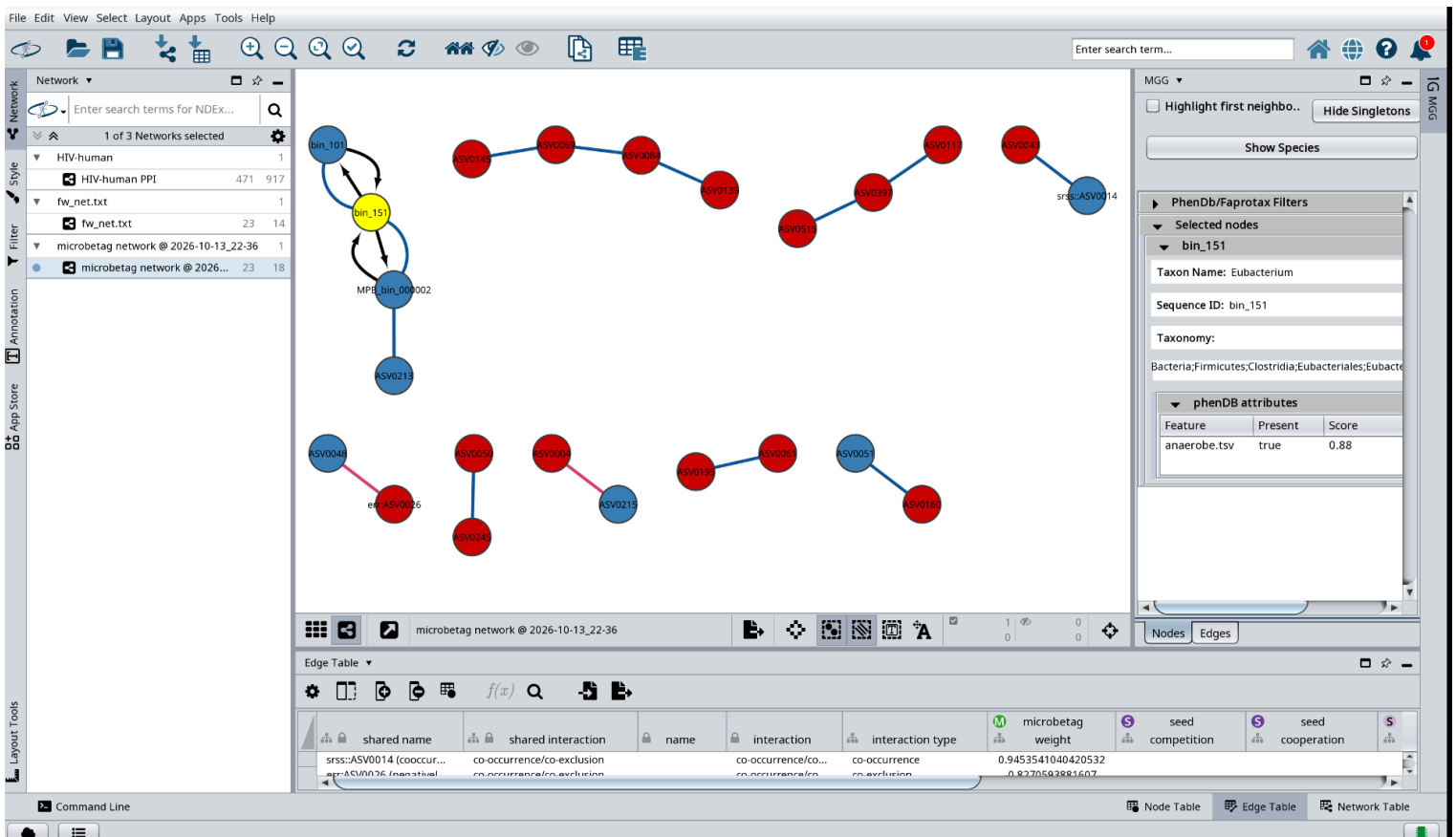



**Edges**: filters, donor → beneficiary




**Edges**: KEGG modules, seed scores

# An annotated network in Cytoscape



# Two ways to run microbetag

▹ On our server

- Against **GTDB v207** genomes
- Submit your **abundance table** or **network** through the MGG Cytoscape app
- Results come back as an annotated network

▹ Locally, with your genomes

- New: **Nextflow** pipeline
- Use your own genomes / MAGs
- Import the annotated network into Cytoscape with MGG

Either way: explore, filter and cluster the annotated network in **Cytoscape**.

microbetag.readthedocs.io · github.com/msysbio/microbetag

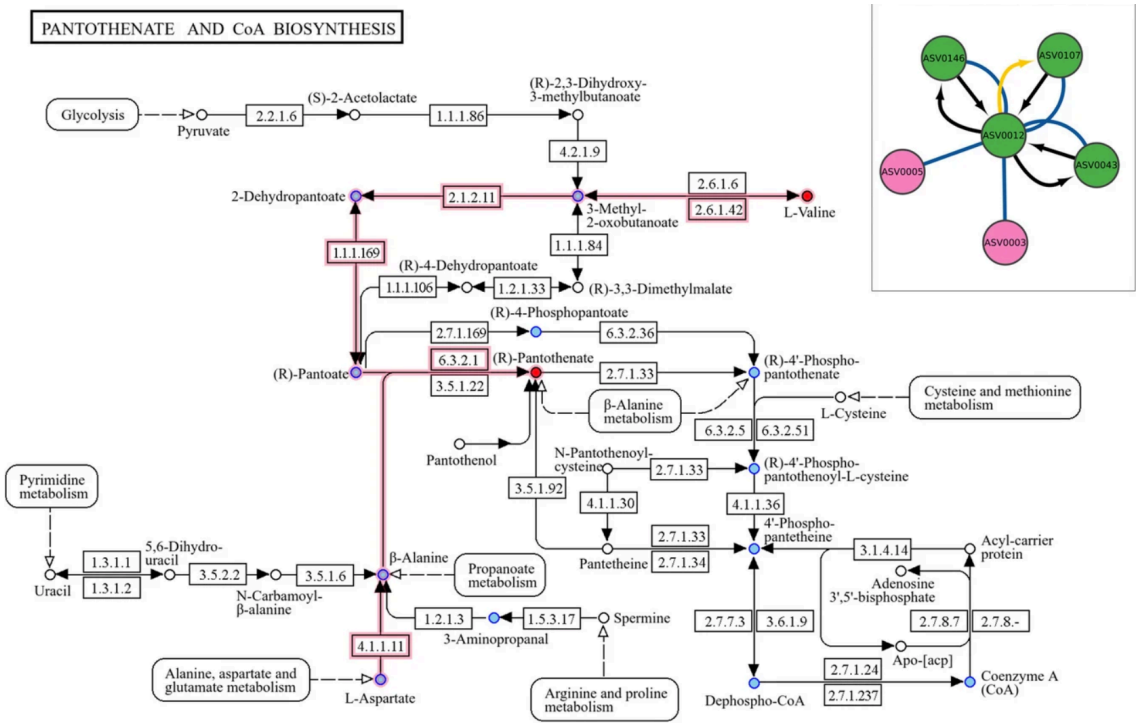
# Case study: infant gut

Cabrera *et al.* (2019)
iron supplementation in infants

*E. lenta* → *F. plautii*
ASV0107 donor ASV0012 beneficiary

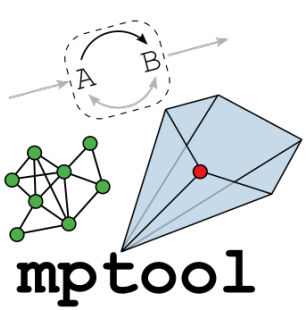
**Complements:** L-valine, (R)-pantothenate
→ vitamin B5 → **coenzyme A**

**Hypothesis:** FOS boosts butyrate & *Flavonifractor* partly via **cross-feeding partners**



# Limits — and what's next

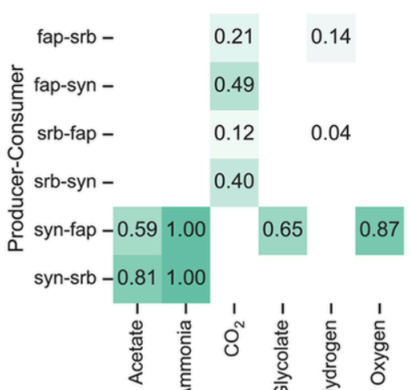
- Hypotheses, not proof: needs experimental validation
- Inherits the network's biases (depth, rare taxa, scale)
- Genome coverage: unmapped taxa stay unannotated
- Next: **interactions' dynamics** — do predicted interactions show up under different media ?




+




→



# Take-home

▹ microbetag

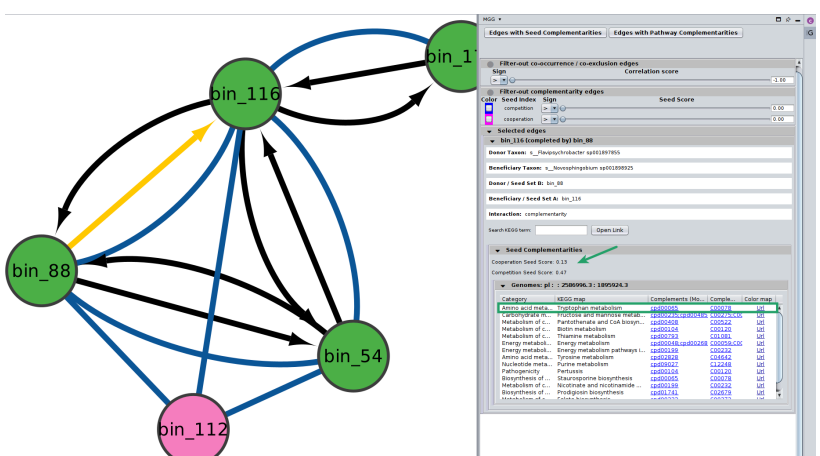
Makes sense of the hairball: from **who is with whom** to testable hypotheses of **what they do and exchange**.

Cytoscape app

Read the docs

Issues & feedback (Matrix)

Read the paper



# Acknowledgements




**Prof. Karoline Faust**
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KU LEUVEN



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NRNB



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KU LEUVEN

ETH zürich **testing the software**



**Aline Schneider**



**Annelies Geirnaert**



**Andrea Erega**

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# Thank you for your attention!

Get these slides

microbetag on GitHub

I'd be happy to take your questions – or get in touch later 👋

charalampos-zafeiropoulos@uni-jena.de

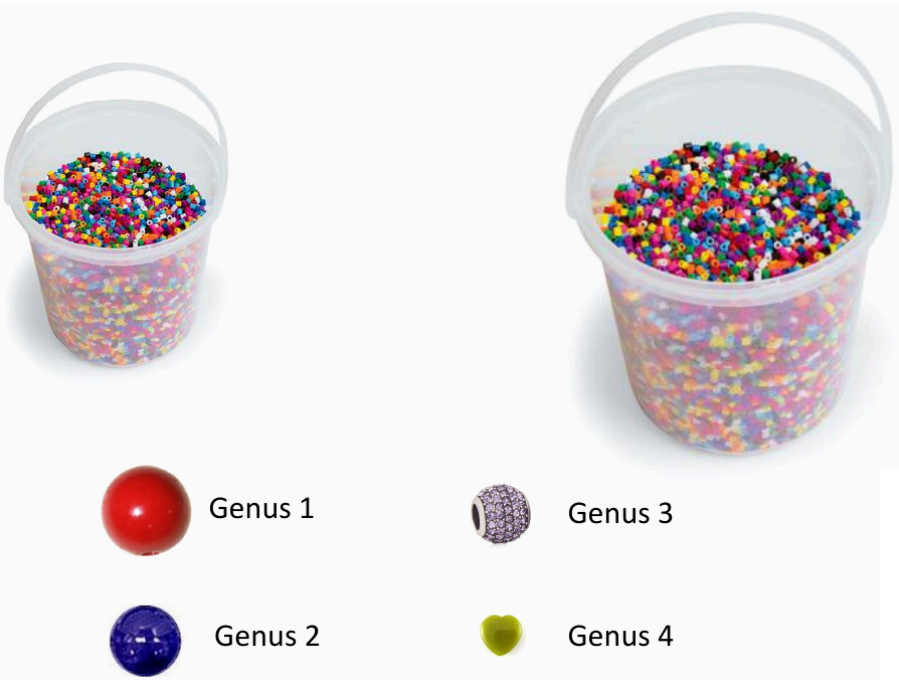
hariszaf.github.io

# Challenges in co-occurrence inference

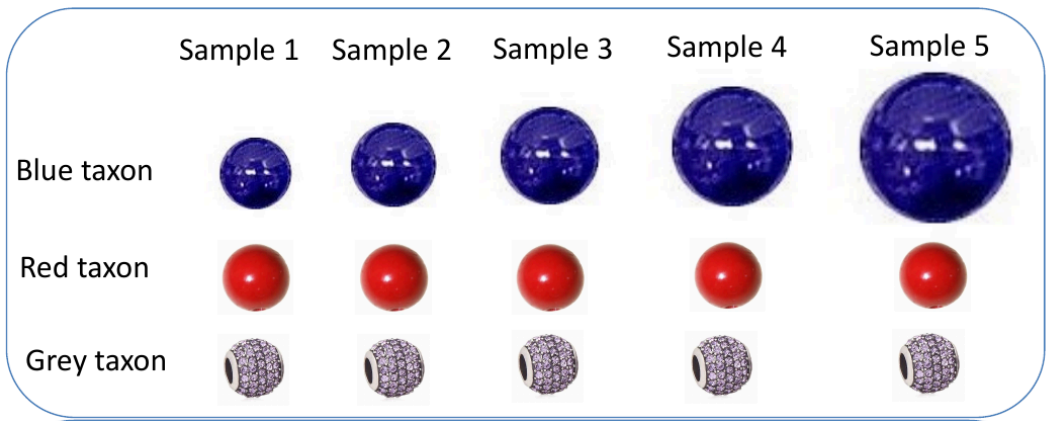
1. **Sequencing depth** varies across samples → compositional artefacts
2. **Rare taxa** → sparse counts, unstable correlations
3. **Indirect edges** → A–C may only reflect A–B and B–C; correlation ≠ interaction
4. **Undirected edges** → who affects whom? an edge has no direction
5. **Sampling scale** → interactions happen at the **microscale**, but we sample at the **macroscale**

# Sampling scale

Technical variability.
Read count ≠ cell count



# Compositionality




**Absolute abundances**

**Relative abundances**